

Sri Hari Sirisipalli

Lead Engineer & AI Systems Engineer · End-to-End

Visakhapatnam, India · sriharisirisipalli@gmail.com · linkedin.com/in/sri-hari-sirisipalli · github.com/srihari-sirisipalli · srihari-sirisipalli.github.io · calendly.com/sriharisirisipalli

Summary

Lead Engineer and AI Systems Engineer. I lead, engineer, and build end-to-end. AI systems, computer vision, GenAI apps, and industry software from concept to production. Digital twins, CFD, FEA, and reliability engineering where the work needs deep physics. Cloud, DevOps, and product delivery to ship it. Marine, offshore, and naval engineering when the project sits in that domain. Fractional CTO at Openct.

The bet the practice is built on: engineering depth and product delivery are more useful together than apart. No handoff between the person doing the analysis and the person doing the app. Same engineer, both halves.

Experience

Fractional CTO, Openct

Feb 2026 to Present · Self-employed · Remote

- Embedded as fractional CTO leading product audit, new feature development, complete openct.co website redesign, backend architecture rebuild, and Flutter mobile app rebuild.
- Complete redesign of the openct.co website with new design system, information architecture, and content flow.
- Rebuilding the Openct Flutter mobile app end-to-end with new architecture, state management, and design system.
- New backend architecture from scratch on DigitalOcean, Redis, and PostgreSQL. Third-party integrations across payments, communication, and analytics.

- CI/CD for mobile and backend. Observability, incident response, and deployment automation.
- Owning architecture decisions, hiring input, code review, engineering standards, and product-engineering delivery cadence.

Lead Engineer & AI Systems Engineer, Independent Systems Engineering

Jun 2023 to Present · Freelance · Remote

Independent engineering practice building end-to-end systems across AI, digital twins, apps, backend, cloud, computer vision, and core engineering. Also leading engineering teams on marine, offshore, and naval design projects.

AI, ML & GenAI

- RAG systems, embeddings, semantic retrieval, and offline LLM stacks.
- Real-time offline 3D AI avatar (deployed): 75% STT and 52% TTS latency reduction. Whisper, Ollama, LangChain, WebSockets, Three.js.
- DocForge: offline document intelligence for engineering and compliance teams.
- Ship Recycling Expert System: HKC, Basel Convention, EU SRR, ILO compliance, 204-activity workflow DAG, 7 PDF report types.

Computer Vision

- Multi-view cattle biometric retrieval at scale: 100,000+ images, SimCLR contrastive embeddings, FAISS multi-index, 20 distributed worker processes.
- Currently building end-to-end computer vision for SCCL Mining Operations (upcoming): 11 detection features and live NOC dashboard.

Digital Twins & Simulation

- Offshore riser fatigue and motion digital twin. Predicted significant wave height at R squared 0.9992 across 88,560 simulated sea states and 40,000+ ML experiments.
- Mooring digital twin, Battery digital twin, Ship emergency-response digital twin.
- MOSES sea-transport DOE automation platform: stratified Latin Hypercube Sampling over 4 ballast zones producing 15,000 DOE cases from 80 input parameters; per-case 12-sheet Excel workbooks (6-DOF motions, structural loads, intact and damage stability across 28 compartments).
- NavalArch Studio: ship design and stability suite with 13 domain subpackages, roughly 450 tests, validated against Maxsurf (4,759 t displacement match).

- PyNAOS: pure-Python FEA engine for offshore structures, 593 tests at 98.6% coverage, six analysis drivers, custom Skyline LDLT solver.

Reliability & ML for Engineering R&D

- Reliability Analysis Workbench for underwater vehicle propulsion structural reliability. OpenTURNS, DOE, ML surrogates over CFD and FEA.
- Multi-input DC-DC converter surrogate and inverse design: physics-informed network with 5-layer OOD gate, roughly 10 to the 4 inference speedup.
- Vibration diagnostics rule engine (ISO 10816): 13 diagnostic rules, 30+ features, PyQt5 workflow.

Apps, Platforms & Industry Software

- zapstays.in (live production): full-stack hostel and co-living platform, roughly 9,000 LOC PHP 8, 50 pages, 60 API endpoints, 20+ DB tables, real-time chat, Google OAuth, admin power-tools.
- VHop nightlife platform (in build).

Backend & Cloud

- AWS (Lambda, EC2, S3, SQS, IAM), DigitalOcean, Redis, PostgreSQL. Terraform, Docker, CI/CD. Async and multiprocessing pipelines for high-throughput workloads.

Marine, Offshore & Naval Engineering

- Leading engineering teams on ship, offshore, and naval design projects.
- Hydrostatics, stability, seakeeping, damage. Offshore platforms, jack-ups, risers, mooring, gangways. Planning craft, service boats.
- Class work against IMO IS Code, SOLAS, IACS, DNV-GL.

AUV & Underwater Systems

- Leading an engineering team on Autonomous Underwater Vehicle (AUV) design and development. Sub-100m class body-of-revolution vehicle, CFRP pressure hull, vectored thruster, autonomous survey missions.
- Requirements-driven MBSE and V-model with FMEA. Covers mechanical, structural, propulsion, and systems work, from concept and pre-feasibility through vehicle delivery.

Software Engineer, LLM & Cloud Infrastructure, Pangeon

Mar 2024 to Present · Part-time · Remote

- Patent similarity and prior-art LLM system at production scale for Intellipat, one of Pangeon's portfolio startups. Supports 1,000+ daily similarity evaluations.
- Embedding-driven retrieval integrated with LLM semantic scoring for automated prior-art ranking.
- Re-architected compute workloads from EC2 to AWS Lambda, reducing infrastructure costs 30% and improving batch completion speed 5 to 7x.
- Reproducible AWS provisioning with Terraform across accounts for consistent cross-account deployments.
- Async and multiprocessing framework for high-throughput evaluation.
- Custom Terraform GUI for AWS resource management.

AI/ML Advisor, Agritech Startup

Mar 2025 to Aug 2025 · Remote

- Benchmarked commercial (OpenAI, Anthropic) and open-source LLMs (LLaMA, Mistral) for agriculture QA.
- Evaluated bilingual answering (English and Telugu) for Andhra agriculture use cases.
- Designed STT fine-tuning pipeline: audio collection, transcription, diarization, dataset prep.

Data Engineer, Sas2Py

Mar 2023 to Jun 2023 · Full-time

- Led migration of legacy SAS data pipelines to PySpark; achieved over 99% functional parity on validation.
- Modeled job dependency graphs to eliminate scheduling bottlenecks and optimize distributed execution.

Machine Learning Intern, Corteva Agriscience

Jul 2022 to Dec 2022 · Hyderabad · Remote

- Converted TensorFlow and PyTorch models to ONNX for cross-platform deployment.
- Optimized inference via structured pruning and post-training quantization.
- Validated numerical consistency across model formats for reliable production deployment.

Software Engineer Intern, Dojima Networks

Jun 2022 to Dec 2022 · Bangalore · Remote

- Integrated Polkadot ecosystem components for cross-chain interoperability.
 - Implemented cross-chain communication workflows for transaction routing and state synchronization.
 - Deployed Prometheus and Grafana monitoring stack for real-time API observability.
-

Publications

ADMOS: An Adaptive Dual-Mode Operational Swarm for Resilient Asset Defense under Communication Jamming. Research Square (Preprint, under review at Discover Vehicles, Springer Nature). Jul 15, 2026. Authors: Mohamed Hijazy Shazin Hassan, Dandu S. N. V. P. R. Varma, Jaswanth Kumar Bobbadi, Neeraj Sree Mailee Dudaboyina, Narendra Sirisipalli, **Sri Hari Sirisipalli**, Giri Rajasekhar Gunnu. DOI: [gist.science/paper/rs/rs-9957095](https://doi.org/10.21203/rs.3.rs-9957095)

Selected Projects

Openct Product & Platform Rebuild · Fractional CTO · openct.co Complete website redesign, Flutter mobile rebuild, backend on DigitalOcean + Redis + PostgreSQL, third-party integrations, CI/CD, and observability. Feb 2026 to Present.

Autonomous Underwater Vehicle (AUV) Design & Development · Engineering Lead Sub-100m class body-of-revolution vehicle, CFRP pressure hull, vectored thruster, autonomous survey missions. Requirements-driven MBSE and V-model with FMEA. Covering mechanical, structural, propulsion, and systems work from concept through vehicle delivery.

Offshore Riser Digital Twin — Motion & Fatigue Coupled hydrodynamics (Bentley MOSES), FEA (custom Python engine and ANSYS APDL), and reliability. Live twin ingests measured motion and current data, updates fatigue budget in real time, and flags risk components for operations.

DocForge — Offline Document Intelligence Local LLMs and embeddings, no internet round-trip. RAG over technical PDFs, patent corpora, and long specs. Python, LangChain, ChromaDB, React front end.

Reliability Analysis Workbench — AUV Propulsion & Offshore Systems DOE across CFD and FEA runs, ML surrogates, and OpenTURNS reliability layer. Live dashboard for design tweak to failure-probability feedback without waiting for a full solve.

Real-Time Offline 3D AI Avatar (deployed) Fully offline 3D agent with 75% STT and 52% TTS latency reduction. Whisper + Ollama + LangChain + WebSockets + Three.js.

LLM Patent Similarity Search · Pangeon / Intellipat Embedding + LLM semantic scoring; EC2 to AWS Lambda re-architecture; Terraform GUI. Mar 2024 to Present.

Goodhar — Cattle Biometric Identification at Scale Large-scale multi-view cattle biometric identification for a state agritech initiative. SimCLR embeddings, FAISS multi-index, near-real-time retrieval. Currently on hold pending data collection.

zapstays.in — Hostel & Co-Living Platform (live production) Roughly 9K LOC PHP 8, 50 pages, 60 API endpoints, 20+ DB tables, real-time chat, Google OAuth, admin tools, CSV export, audit log.

PyNAOS: Pure-Python FEA Engine for Offshore Structures 6 analysis drivers, 3D Timoshenko beam elements with P-Delta, Skyline LDLT solver, PyQt5 workstation, 593 tests at 98.63% coverage.

NavalArch Studio: Ship Design & Stability Suite 13 domain subpackages, roughly 450 tests, Maxsurf-validated hydrostatics, IMO/SOLAS/IACS/DNV-GL rule integration.

MOSES Sea-Transport DOE Automation Platform 15,000 DOE cases over 80 parameters, PyQt5 orchestrator, XGBoost/PyTorch/LightGBM surrogates, SALib sensitivity, Optuna/PyMoo optimization.

Ship Recycling & Dismantling Expert System Django app; HKC + Basel + EU SRR + ILO compliance; 204-activity workflow DAG; 7 report types.

Multi-Input DC-DC Converter Surrogate & Inverse Design Physics-informed network with 5-layer OOD gate; XGBoost R squared 0.985; roughly 10 to the 4 inference speedup; PyQt5 GUI for parameter pinning.

Vibration Diagnostics Rule Engine 13 diagnostic rules for rotating machinery (unbalance, misalignment, bearing defects, rotor faults); 30+ features; ISO 10816 severity mapping.

Education

B.Tech, Mechanical Engineering · Mahindra University, Hyderabad Aug 2018 to Jun 2022 · CGPA 7.5

Coursework: Linear Algebra & Matrices, Probability & Statistics, Data Structures, Big Data Computing, Advanced Data Analytics, Time Series Forecasting, GPU Programming, Computer Aided Engineering Design.

Activities: Zenith Science Club core team (MU Research Symposium 2020); AERO Sports Meet volunteer (2019 to 2022).

Skills

AI, ML & GenAI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face, Transformers, ONNX, Surrogate Modeling, Physics-Informed NNs, OOD Detection, SHAP, Feature Engineering, Model Optimization.

GenAI & LLMs: LangChain, LlamaIndex, RAG Pipelines, FAISS, ChromaDB, Vector Databases, Embedding Models, Ollama, vLLM, Whisper STT, Prompt Engineering, Offline LLM Stacks.

Computer Vision: OpenCV, YOLO, BoT-SORT, SimCLR, ResNet, Tesseract, MediaPipe, Visual Similarity Search, Multi-View Embeddings, Ranking Fusion.

Digital Twins & Simulation: Digital Twins, CFD (ANSYS Fluent, OpenFOAM, k-omega SST), FEA (ANSYS APDL, ANSYS Mechanical, custom Python engine), Reliability Analysis (OpenTURNS), DOE, Latin Hypercube, SALib, Optuna, PyMoo.

Engineering CAD/CAE: ANSYS Fluent, ANSYS APDL, ANSYS Mechanical, OpenFOAM, Bentley MOSES, Autodesk Inventor, SolidWorks, FreeCAD, Fusion 360, AutoCAD, Maxsurf, Rhino.

Mobile, Web & Apps: Flutter, Dart, React, React Native, Next.js, TypeScript, Vite, Tailwind, FastAPI, Django, WebSockets, HTMX, PyQt5, PyQt6, Streamlit.

Backend & Cloud: AWS (EC2, Lambda, S3, SQS), DigitalOcean, GCP, Redis, PostgreSQL, MySQL, SQLite, Terraform, Docker, CI/CD (GitHub Actions), IAM, Nginx.

Data Engineering: PySpark, Hadoop MapReduce, ETL Pipelines, NumPy, SciPy, Pandas, Polars, Airflow, Prometheus, Grafana, MLflow, Weights & Biases, Data Validation.

Programming & Tooling: Python, TypeScript, JavaScript, Dart, C++, MATLAB, SQL, Bash, Java, Go, C, Git, Linux.

Marine, Offshore & Naval: Naval Architecture, Ship Design, Offshore Structures, AUV, Hydrodynamics, Seakeeping, Mooring, Stability, IMO IS Code, SOLAS, IACS, DNV-GL.

Leadership & Activities

- **CODEIAM Mentor** (Jul 2024): Mentored 3 teams at a university hackathon; 2 teams ranked in top 5 of 40.
 - **AERO Sports Meet** (2019 to 2022): Volunteer and Security team for annual campus events.
 - **Zenith Science Club** (2019 to 2022): Core team for MU Research Symposium 2020.
-